

Seeds of Salvation

A Game-Based Approach to Environmental Awareness for the Young Generation through Storyline and Real-World Environment Challenges.

This project is submitted to the Department of Computer Science and Engineering, Dhaka International University, in partial fulfillment of the requirements of the Bachelor of Science (B. Sc.) in Computer Science and Engineering (CSE).

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Declaration

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Supervisor's Statement

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**Dedicated to
Our Parents
&
Teachers**

Abstract

In this era of the internet and technology, video game is a leading medium among entertainment world. It is not just considered as only an entertainment medium, but a complex, technology-driven medium that combines art, storyline, simulation, and interactivity. But this entertainment medium brings excessive screen time and reduces real-world awareness, leading to addiction among teenagers. One study shows that 57.32% students are mildly addicted, and 28.57% are severely addicted in Bangladesh. On the other hand, the environment is getting worse day by day; deforestation, air pollution, water pollution, soil pollution, plastic pollution, biodiversity loss, etc. causing global warming. Although many organizations are working to increase environmental awareness, despite all their efforts, they are not having much impact on teenagers. This is why this project is working to make a solution to those two problems by using our teenagers' bad habits to increase environmental awareness within them. This project uses Unreal Engine 5 to develop a third-person video game, which will give a visual perception on environment pollution. Environmental Awareness, Behavioral Influence, and Sustainable Learning are the main objectives of this project. Besides the Mobile and PC Game Education Institute, Non-Governmental Organizations (NGO) also has its application to increase awareness. Overall, Seeds of Salvation has the potential to make a positive influence among the teenagers and educate the next generation about their responsibilities on environment.

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1.1 Overview

In the 21st century, known as the Digital Era, the internet and technology are deeply integrated with human life. Modern technology and the internet have changed human life dramatically and are influencing every aspect of it. From communication to education, business, and entertainment, modern technology has reshaped every aspect of human life. This integration of technology has brought many remarkable achievements, like solving complex problems, enhancing connectivity, and creating new opportunities for innovation and creativity.

Among the many technological achievements, video gaming has emerged as one of the most influential and fastest-growing industries. Although initially, video gaming was developed for entertainment, but now evolved into a complex, technology-driven medium that combines art, storyline, simulation, and interactivity. Now, modern games are not just a pastime, but they serve as a platform of learning, social connectivity, and even environmental and psychological awareness with the help of powerful game engines, artificial intelligence, and virtual reality.

1.2 Problem Definition

Video games are one of the most delightful forms of entertainment among teenagers. This often results in excessive screen time and reduces real-world awareness. They spend time excessively playing games and forget what is happening around them. This causes increasingly bad habits, mental illness, promotes social isolation, etc. This is a very concerning fact for our teenagers and young adults. They are getting weak due to an unhealthy routine, not taking meals on time, not getting to bed in time, and falling behind in education. One study shows that 57.32% students were marked as having mild addiction, and 28.57% had severe addiction in Bangladesh [1].

On the other hand, our planet is facing critical environmental challenges. Deforestation, air pollution, water pollution, soil pollution, plastic pollution, biodiversity loss, etc. causing global warming. The ozone layer is depleting, and the sea level is rising. In this situation, many organizations and societies are defending this with various activities, increasing awareness. Despite all their efforts, they are not having much impact on teenagers. The teenagers are very indifferent to thinking and acting for the environment.

1.3 Objectives

This project is trying to solve those problems in an engaging way that the teenager can easily adapt. That is how this project can bring some value to society. The main objectives of this project are:

- **Promote Environmental Awareness**

The project aims to increase environmental awareness among the players and educate them about real-life environmental problems through storyline and challenges. By embedding the results of pollution, deforestation, and climate change into the virtual realistic game, players will be directly affected by the results. That will make them more aware and sincere about the environment.

- **Behavioral Influence**

This Project will influence players' thoughts on the environment and motivate them to make changes in the real world. Through the meaningful gameplay, decision based consequence, and realistic output player will witness their actions, which will bring positive effects on their behavior.

- **Sustainable Learning**

The project provides a sustainable and engaging way to educate on scientific facts about the environment and motivate players to care about the environment. By integrating various scientific data into the gameplay, storylines, and challenges, players will be able to understand complex environmental facts.

- **Interactive Community**

The project aims to build an interactive community where players can share and interact with others' eco-friendly ideas and compete with each other in environmental challenges to make changes. By encouraging collaboration and healthy competition, all players will be able to practice sustainable learning on environment.

- **Research Integration**

A scope of research on players to evaluate the psychological and behavioral effects of handling real-life environmental challenges in the game. Researchers can analyze players' data and find out the most efficient way to change the environment in the long term.

1.4 Motivations

The motivations of this project are:

- The social problems are the main motivation. The young generation is addicted to smartphones. They are spending countless hours scrolling through social media or playing games. And the sad thing is that those have no educational or moral values.
- At the same time, environmental degradation is increasing at an alarming rate. Many organizations are working, having good results, but have failed to attract our teenagers' attention.
- This project's main goal is to combine two problems into one solution, using the power of video games, to make environmental awareness more fun, engaging, challenging, and rewarding. This project is trying to bring good from the bad, using the teenagers' bad habit of gaming to teach them how to care for our environment.

1.5 Applications

Beside as Mobile and PC game, this project has many fields of application:

- **Environment Awareness Program**

Government agencies and non-governmental organizations (NGOs) can use this game as a part of their public awareness campaign aiming the teenagers and young adults. This interactive game can deliver an impactful message to the people.

- **Educational Institutions**

The project can be used as an interactive learning tool in educational institutions like schools, colleges, and universities to support environmental science. By integrating gameplay with scientific facts and real-world environmental scenarios, the students can understand complex concepts such as climate change, pollution, and ecosystem balance more effectively than traditional classroom methods.

- **Youth Engagement Community**

The project can serve as a digital medium to attract young users; whose who are already active in gaming communities.

- **Research and Behavioral Studies**

Researchers can use this platform to study how this game influences the behavioral change of teenager player. Which will help to improve the promotion strategy of actual environmental awareness.

- **Cross-Platform Deployment**

The game can be deployed as a Mobile, PC, and Console Game to ensure global access.

2.1 Overview of the Video Game Industry

The video game industry is one of the most fast growing industry in the global entertainment market. Over the past few decades, it has evolved from the simple pixel-based game to a complex, realistic, cinematic game with advanced graphics, music, and online connectivity. Even Artificial Intelligence is also including into this industry to make every gameplay more challenging and more realistic. Modern video games are developed for multiple platforms like personal computers, gaming consoles, and mobile devices. Which ensures reaching the global audience. The industry covers a wide range of gaming, such as action, simulation, creative, educational, strategical etc.

Technical achievements have played a major role in shaping this industry. With modern game engines like Unreal Engine and Unity, developers can create a more realistic experience. The industry includes development, marketing, distribution, and monetization. Which are increased dozens of job disciplines and thousands of job opportunities worldwide. With continuous innovations, expanding markets, and increasing users, the video game industry has become a powerful culture, influencing digital life.

2.2 Problem Study

• Video Game Addiction

Video game addiction is one of the most concerning topics for parents nowadays. One study shows that, among 1125 Bangladeshi university students, more than half of the students (54.3%) were classified as severely addicted [2]. Another study shows that among 502 school and college students, 57.32% students were marked as having mild addiction, and 28.57% had severe addiction in Bangladesh [1]. This research found several health factors like loneliness, anxiety, headaches, eye discomfort, and blurred vision among those teenagers. Another one found children parent troubled relation, emotional debouchment and social isolation [3].

• Environment Pollution

On the other hand, a study shows that PM2.5 and PM10 exceed national and WHO air quality standards in Dhaka, Bangladesh. The pollution concentration is 5-6 times higher than the safer limits [4]. Like this case, other elements like water and soil are also contaminated with harmful chemicals and heavy metals, causing harm to human life [5]. Another public health study shows that 90% of respondents experienced health effects from

air pollution, 60% from water contamination, and 55% from noise pollution in Dhaka, Bangladesh [6]. This is the condition of the environment, and the teenagers are not completely aware of that.

- **Lack of Environmental Awareness Among Teenagers**

Recent research on environmental awareness among the young generation in Bangladesh reveals a significant gap between theoretical and real or practical knowledge on environmental issues. Research shows that environmental education in Bangladesh is insufficient to develop a genuine environmental awareness and behavior change among students. Those students' awareness level on environment is very low. The notional education framework does not comprehensively address the environmental issues in textbooks or curriculum completely, resulting lack of awareness among students. Overall, those study shows there are a lot of gaps to fill to increase the environmental awareness among teenagers [7].

2.3 Related Works and Comparison

There are many environment-related games, but mostly they are focused on survival. So after a thorough inspection, we selected four games that promote environmental awareness. Those are:

- **Plasticity**

It is a semi-3D, puzzle-based, cartoonistic game about a plastic-polluted world. The player has to solve puzzles and clean plastic plastic-polluted environment through the gameplay. Published in 2019 and created by 30 students of Plasticity Games [8].

- **World Rescue**

It is a research-based game inspired by the goals of the United Nations. It is a 2D, storyline-based game that promotes displacements, disease, deforestation, drought, and pollution. Published in 2015 by UNESCO [9].

- **Elva the Eco Dragon**

It is a 3D game. In this game, the players' main mission is to clean the planet by collecting and recycling waste, planting trees, cleaning up oceans, etc. It was published in 2020 by Titan Deep Space Company [10].

- **Terra Nil**

It is a strategy-based, semi-3D, cartoonistic game. Where player's main objective is to restore the environment from a wasteland using technology. Published in 2023 by Devolver Digital [11].

Table 2.1: Comparison of these games with this project (Seeds of Salvation)

Name of Games	Contribution Field	Graphics and Gameplay	Weakness	Scope of Improvement
Plasticity	Plastic pollution	semi-2D, cartoonistic,	Graphics and Reality	Real-life problem implementation
World Rescue	Deforestation, Drought, Pollution	2D, cartoonistic	Graphics and Reality	Real-life problem implementation
Elva the Eco Dragon	Recycling waste, Planting trees, Cleaning up oceans	3D, cartoonistic	Reality	Real-life problem implementation
Terra Nil	Restore wasteland	semi-2D, cartoonistic	Graphics and Reality	Real-life problem implementation
Seeds of Salvation	Restore the environment	3D, realistic, open-world	-	AR and VR

3.1 Overview

According to the standard game development method, this project also started with the planning phase. After careful planning and designing, this game achieved its Game Objective and Game Progression. Through the step-by-step workflow, this project maintained the standard of Game Development Life Cycle (GDLC).

3.2 Concept of Game Development Life Cycle (GDLC)

Game Development Life Cycle (GDLC) is a standard framework guide of game development from its initial planning to the post-production phase. It ensures both technical quality and an engaging player experience. It includes a total of six different phases.

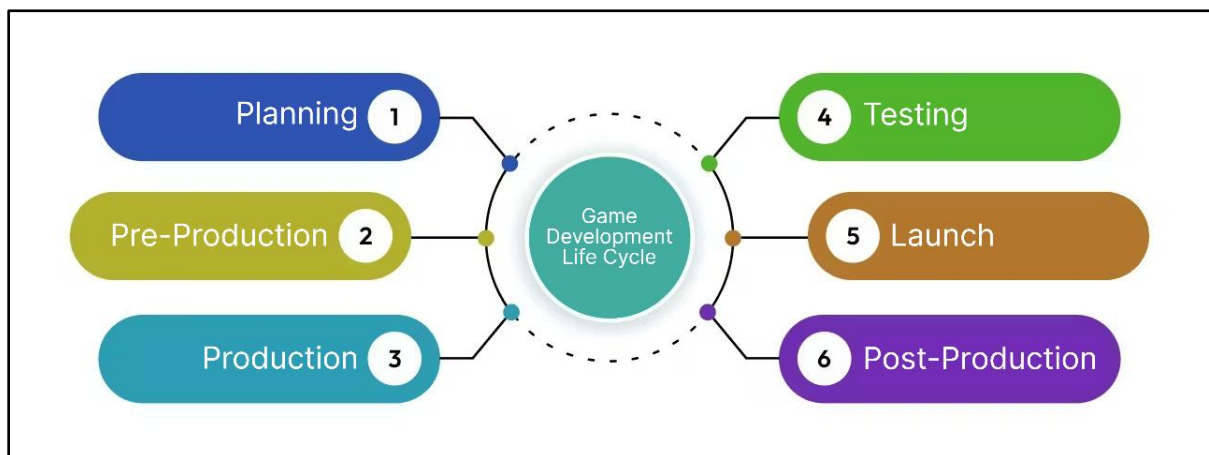


Figure 3.1: Game Development Life Cycle (GDLC)

- **Planning**

In this phase of game development, the objective is to define the core concepts, genre, and target audience. Also, establishing the game objective and breaking down the task of the entire project.

- **Pre-Production**

Creating a blueprint of story, game mechanics, and game design. Also, prepare a complete documentation of the project blueprint.

- **Production**

This is the main development part of the project. Where all assets are designed, and code is written to build the game.

- **Testing**

This is the quality assurance phase. Build alpha and beta versions of the game, then find and fix the bugs. Ensuring all functionality works properly without any issues.

- **Launch**

In this phase, there are two tasks. First, pre-launch or the marketing of the game to reach the maximum audience. Then release the final version of the game to the market.

- **Post-Production**

This phase is mainly monitoring and maintenance of the server and the game. Releasing the patch and fixing cross device bug based on the feedback of the players. Also, releasing new features and new updates to the gameplay are part of this phase.

3.3 Unreal Engine

This project is using Unreal Engine 5 (UE5). Unreal Engine revolutionizes game development and visual production with its unique core features. Each feature helps to create unimaginable, ultra-realistic games [12]. The key features and advantages of Unreal Engine 5 are:

- **Rendering and Lighting Features**

- i. **Nanite**

- Nanite allows developers to work with a massive amount of geometry details with the micropolygon geometry system. Developers can easily handle assets with millions of polygons without any performance issues.

- ii. **Virtual Shadow Maps (VSMs)**

- It works with Nanite to produce real-time, consistent, higher-resolution shadows. Which makes a scene more realistic.

- iii. **Lumen**

- It is a fully dynamic lighting solution that reacts immediately with scene changes and generates real-time reflection to create a realistic scene.

- iv. **Virtual Texture**

- It uses two methods to handle texture, dividing them into small tiles. Which improves GPU performance for loading graphics.

- **Simulation and Effects Features**

- v. Niagara**

- Niagara creates film-quality VFX in real-time, like fire, smoke, dust, and water. It allows developers to use a complex, customizable particle system. Which reacts with motion, particle-to-particle communication, and the environment.

- vi. Chaos Physics**

- It is a next-generation high-performance physics system. Which allows to create destruction and shattering scenes. Chaos also supports mesh dynamics, cloth, and hair interaction with effects from dust and smoke with Niagara.

- **World Building Features**

- vii. Built-in Modeling Tools**

- There are extensive tools for mesh editing, dynamic sculpting tools, geometry script in the editor, UV creation, etc.

- viii. World Partition**

- This built-in function allows to create open world by automatically dividing the world into a grid cell system.

- **Character and Animation Features**

- ix. MetaHuman**

- This allows developers to create and animate a complete digital 3D human with ultra-realistic face and body properties. Also allows to create realistic emotions to face mesh with Face DNA.

- x. Runtime Character Animation**

- This animation tool allows to make animation sequence at runtime for different gameplay scenarios. Also allows adjusting character pose with root motion, for example, walking, running, climb wall, etc.

3.4 Game Objectives

This project follows an unpublished novel named 'Seeds of Salvation' as its storyline. According to the storyline earth environment was uninhabitable for the past 300 years. After 4 generations in space, a curious 21-year-old botanist decided to come to Earth to restore the environment. He descend earth in a certain region and began planting trees. Planting trees, clearing pollution, and restoring the environmental balance are the main objectives of this game.

3.5 Game Progression

This game follows the storyline as its main progression. Its progression flow chart:

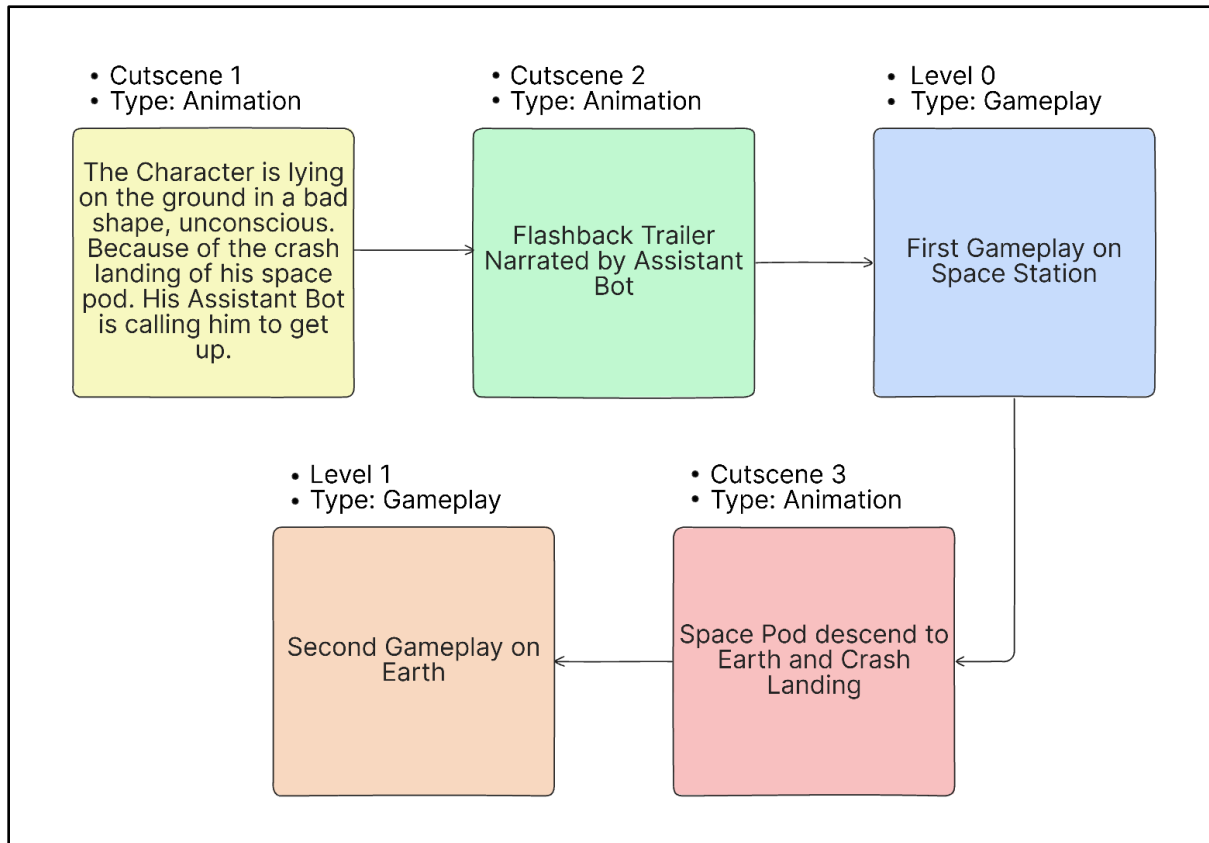


Figure 3.2: Game Progression

3.6 Workflow

This project was prepared for the purpose of fulfilling the requirements for the Bachelor of Science (B.Sc.). So it's not a complete game development at the market publishable level. This project covers and restructures all steps into three main phases. Those phases are:

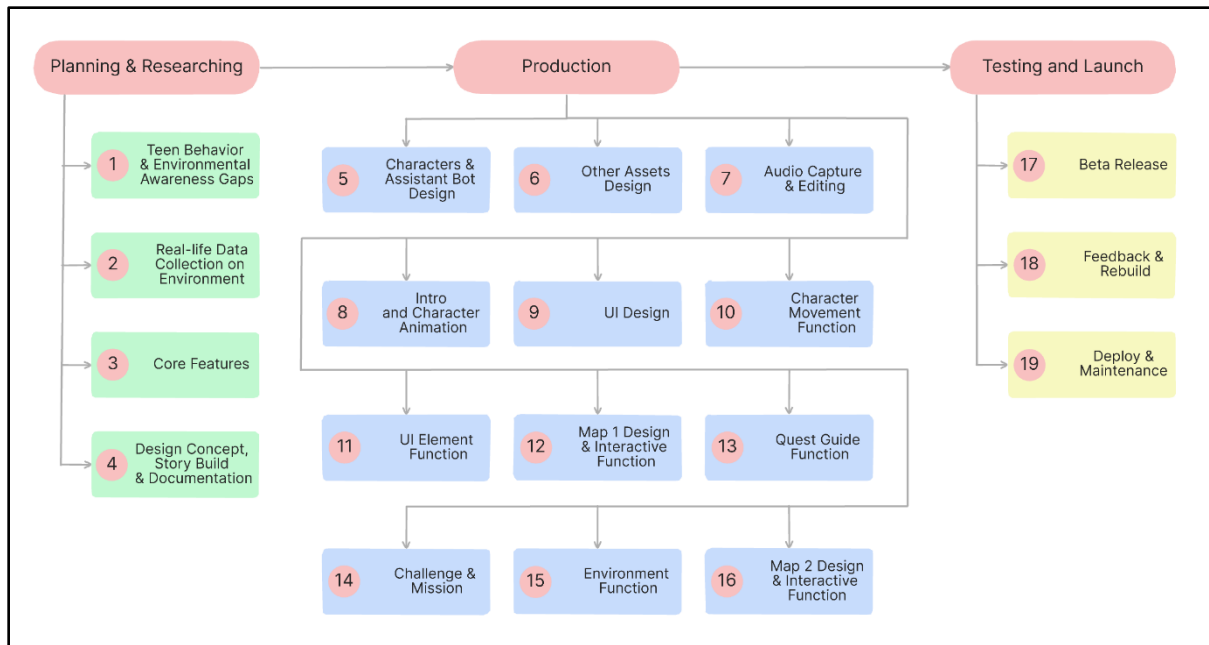


Figure 3.3: Project Workflow

• Planning and Researching

Research on teenagers' behavior and environmental awareness gap is one of the main steps in the planning phase. Also, real-life environment data collection, core features design, design concept, etc., steps have been completed in this phase. This is the main preparation phase of the project.

• Production

This is the main design and development phase of the project. All design-related steps of this project, including character, other assets, map, and UI design, were completed. Also, programming steps like character movement function, UI element function, and interactive function were completed in this phase of the project

• Testing and Launch

First, releasing an alpha or beta version, then rebuilding the project according to feedback, is crucial for game development. That's why beta release, feedback collection and analysis, rebuild, and deployment were the part of this phase.

3.7 Planning and Research Steps

This phase includes four steps of this project. Those are:

i. Teen Behavior and Environment Awareness Gaps

In this step, research and analysis were conducted to understand the problem set of this project. The focus was on how to improve teen awareness of environmental issues.

Insights from this analysis were used to design the realistic learning objectives, game mechanics, and narratives

ii. Real-life data collection on environment

In this step, some real data were collected, such as oxygen and carbon dioxide production rate, nutrition absorption rate, energy consumption rate of a tree, etc. Also percentage of different types of gas in the air, human oxygen consumption rate were found out in this step.

iii. Core Features

In this step, the core features of this project were defined based on the game objectives. This game is a third-person game that ensures a realistic game experience. Unreal Engine 5 was selected because of its advantage features. Also, map type, challenges, and technical terms were sorted and defined.

iv. Design Concept, Story Build, and Documentation

In this step, the overall design concept of the project was defined. Visual style, user experience, and thematic consistency with the project objective are also defined in this step. Making game objectives, game progression, and documenting all those plans were the last step of the planning phase.

4.1 Overview

This phase focuses on the comprehensive development of virtual, audio, and functional components. Those components bring the game world to life with an engaging gameplay experience. It includes the design and implementation of characters and various in-game assets, along with capturing and editing audio. This phase also includes the user interface and its functionality, ensuring smooth navigation and interactive gameplay. Overall, this phase serves as the backbone of this project, integrating artistic design with a functional system to create interactive, realistic gameplay.

4.2 Development Steps

This phase includes fifteen steps of this project into two categories. Those are:

- **Production**

- i. Characters & Assistant Bot Design**

This step focused on creating the primary game characters and an intelligent assistant bot that supports gameplay progression and environmental education. This step involved designing character appearances, personalities, and roles that align with the storyline. Two MetaHuman characters were created using human anatomic data with custom clothing as the storyline.

An assistant bot was designed to act as a guide, providing instructions, hints, environmental facts, and mission-related feedback to players. This bot plays a major role in educating players, explaining real-life environmental issues, the consequences of player actions, and eco-friendly practices in an interactive manner.

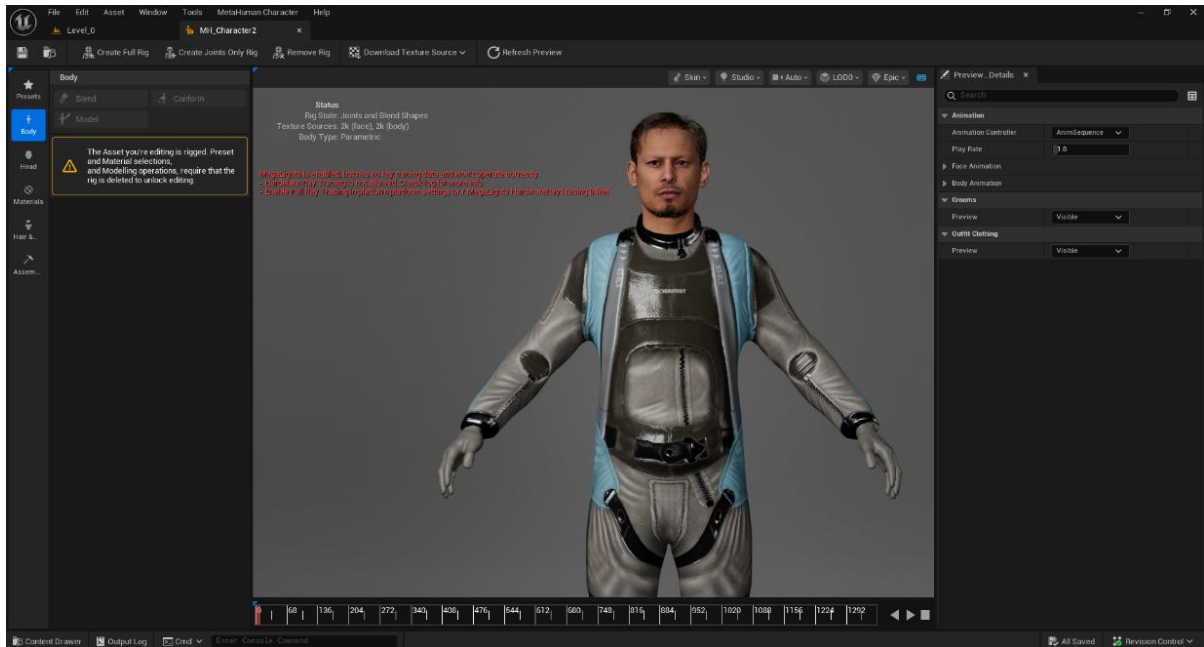


Figure 4.1: MetaHuman Character (Full Body)

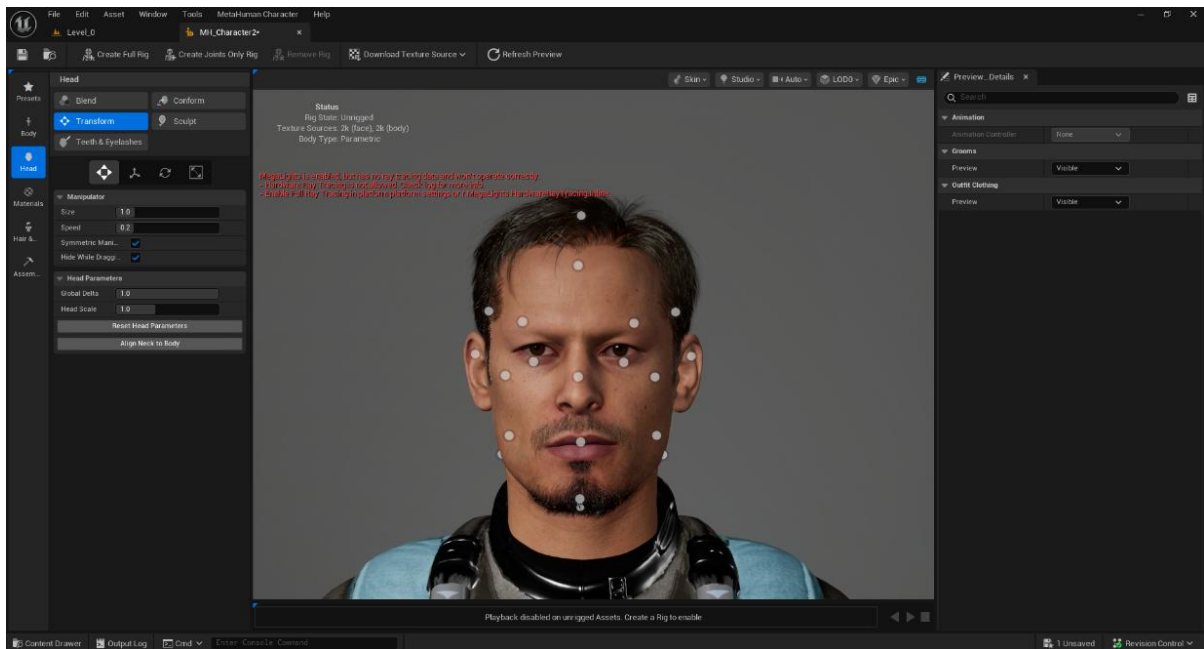


Figure 4.2: MetaHuman Character (Facial Control Rig)



Figure 4.3: Assistant Bot

ii. Other Assets Design

This step involves designing and supporting the environment and interactive objects such as trees, tables, chairs, doors, and other props that decorate the game world. Each asset was modeled and textured to match the storyline and style of the gameplay. Overall, this step is the initial preparation to design an actual game level.

iii. Audio Capture & Editing

This step focused on developing high-quality sound that enhances emotional impact and realism in the gameplay. For the limitation of a Sound Engineer or skill set, this project uses Eleven Lab's Eleven V3 AI model to generate speech and sounds. This model allows emotional expression voice to generate speech.

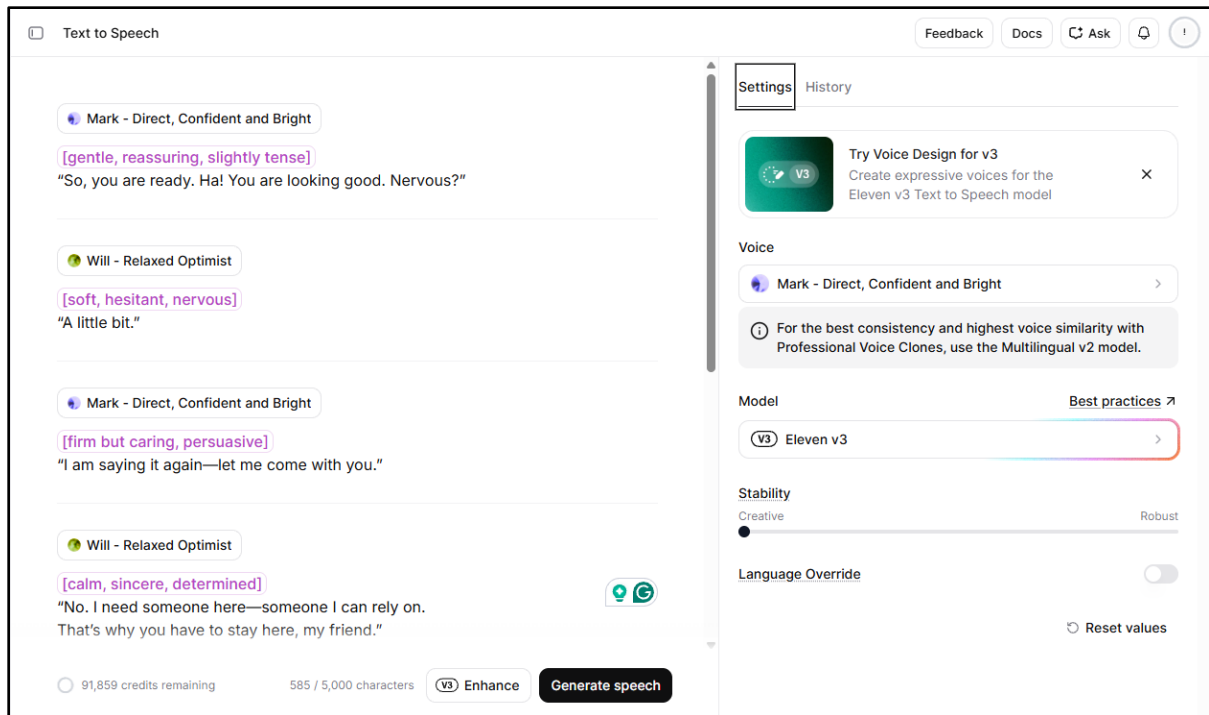


Figure 4.4: Eleven Lab's Eleven V3 AI Model

iv. Intro and Character Animation

This step focused on bringing the narrative and character to life through cinematic introduction and realistic character movement. Intro is the flashback story of the game progression. On the other hand, the Unreal Character Control Rig and IK Rig were used to animate a character.

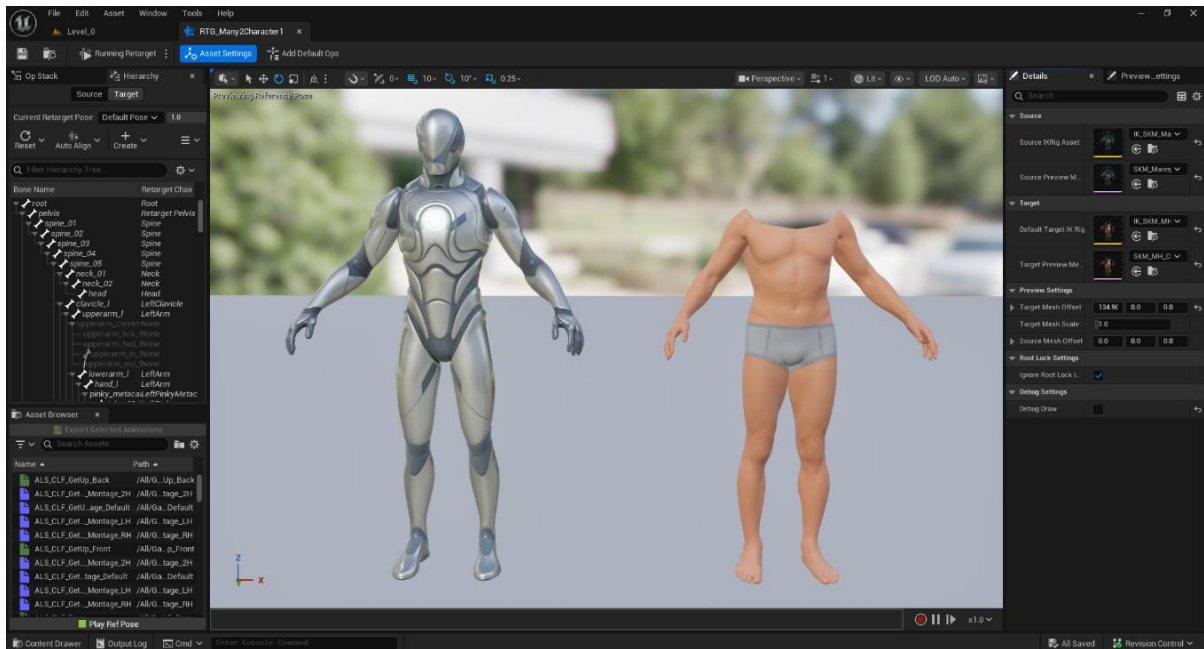


Figure 4.5: Unreal Character Animation Rig

v. UI Design

This step focused on creating a clear and visual constant user interface that supports smooth gameplay and interaction. UI layouts were structured to present information clearly, while maintaining the color scheme, typography, and iconography to enhance readability. Overall, this step ensures that the user interface effectively supports gameplay and storyline.

vi. Character Movement Function

This step focused on implementing smooth and responsive character movement. The core movement like walking, running, and jumping, were developed in precise, natural manners. Overall, this step ensures that the character movement supports environmental interaction, narration, and progression effectively.

vii. UI Element Function

This step focused on implementing the functionality of the user interface. This is the programming part of the UI Design step. During this step, interactive elements such as buttons, menus, sliders, pop-ups, mission tracker, and notification panel were programmed. Overall, this step ensures that the UI works properly within the gameplay.

viii. Map 1 Design & Interactive Function

This step focused on designing and implementing the first playable level in the game. According to the storyline, it is a space station. In the game progression, it was marked as Level 0. Also, interactive functions were integrated into the map that allow players to engage with objects like doors, chairs, etc. Trigger, collision system, and event-based action were implemented to guide players.

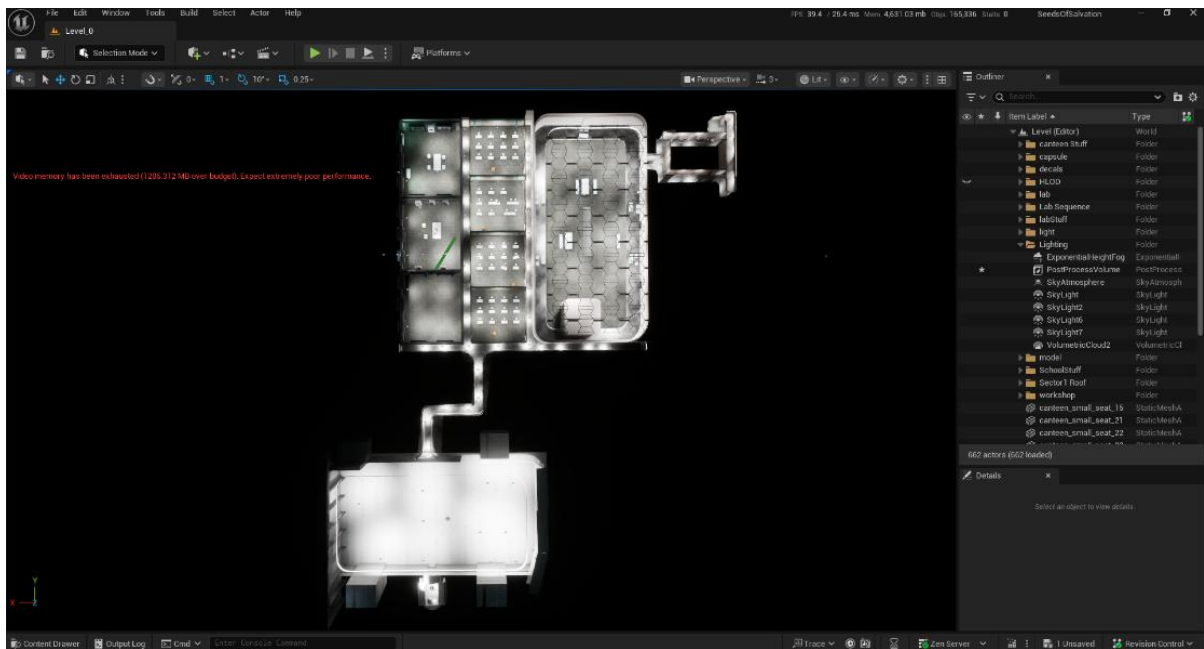


Figure 4.6: Map 1 (Top View, Space Station)

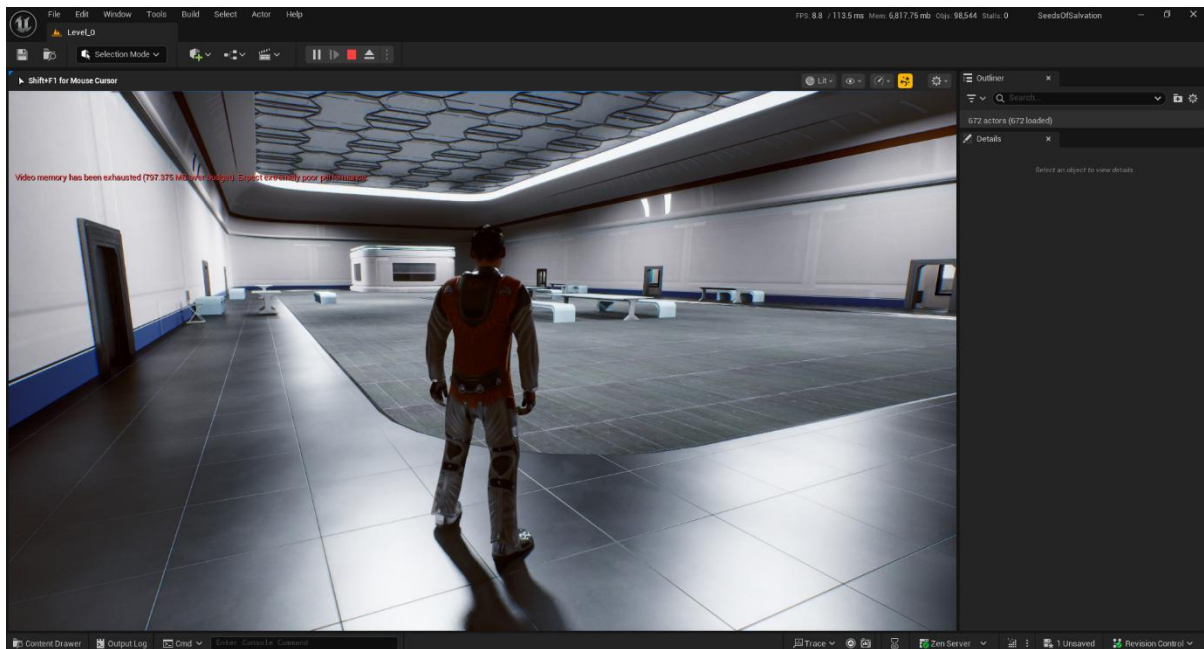


Figure 4.7: Map 1 (Cafeteria, Space Station)

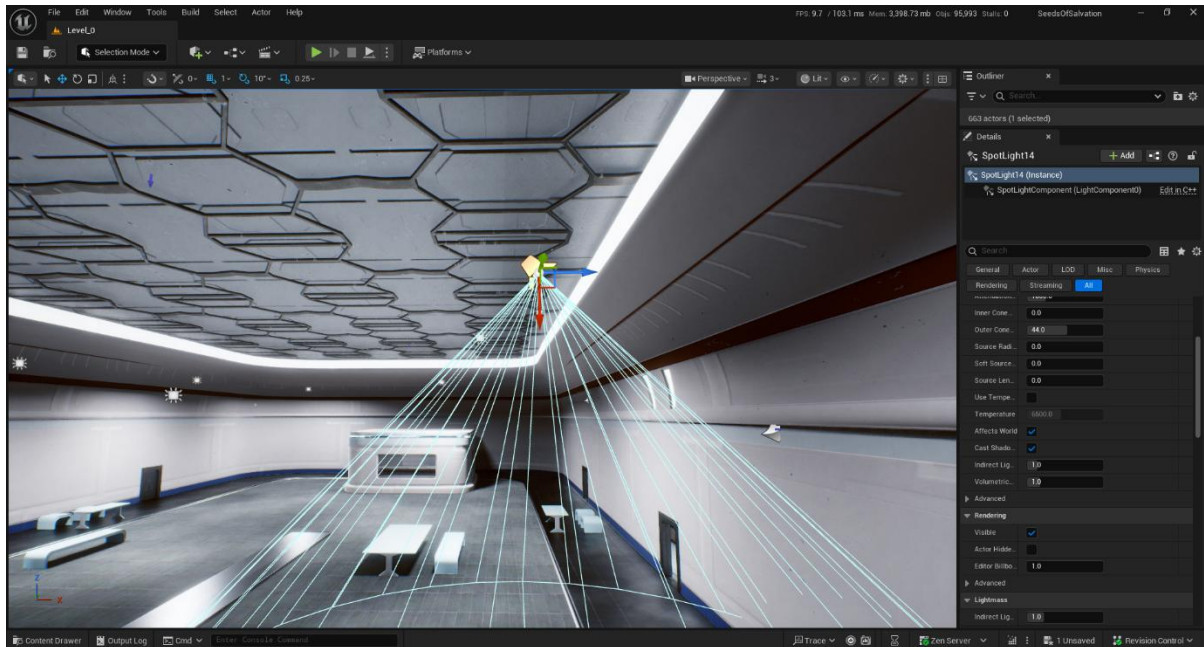


Figure 4.8: Controlling Light with Lumen

ix. Quest Guide Function

This step focused on developing a quest guide system to guide the players to achieve objectives and challenges. This quest guide function ensures a smooth gameplay flow, reduces player confusion, and supports game progression.

x. Challenge & Mission

This step focused on designing and implementing missions and challenges into the gameplay. Various missions and challenges were implemented align with the storyline and objective of the project. Overall, this step ensures that missions and challenges serve as the primary mechanism for learning and connecting to the gameplay.

xi. Environment Function

This step focused on creating a dynamic function that can reflect the real world eco system. Calculating air pressure, oxygen production, and consumption rate, temperature, etc. factor are calculated and integrated in real-time within the gameplay. Overall, this function ensures the implementation of a realistic environment to achieve this project's objective.

xii. Map 2 Design & Interactive Function

This is the second playable level in the game. According to the storyline, this is a certain region on Earth. It is a real-life-like environment with soil, river, trees, etc. In the game progression, it was marked as Level 1. Overall, this level ensures make an impact on the players on environmental awareness.

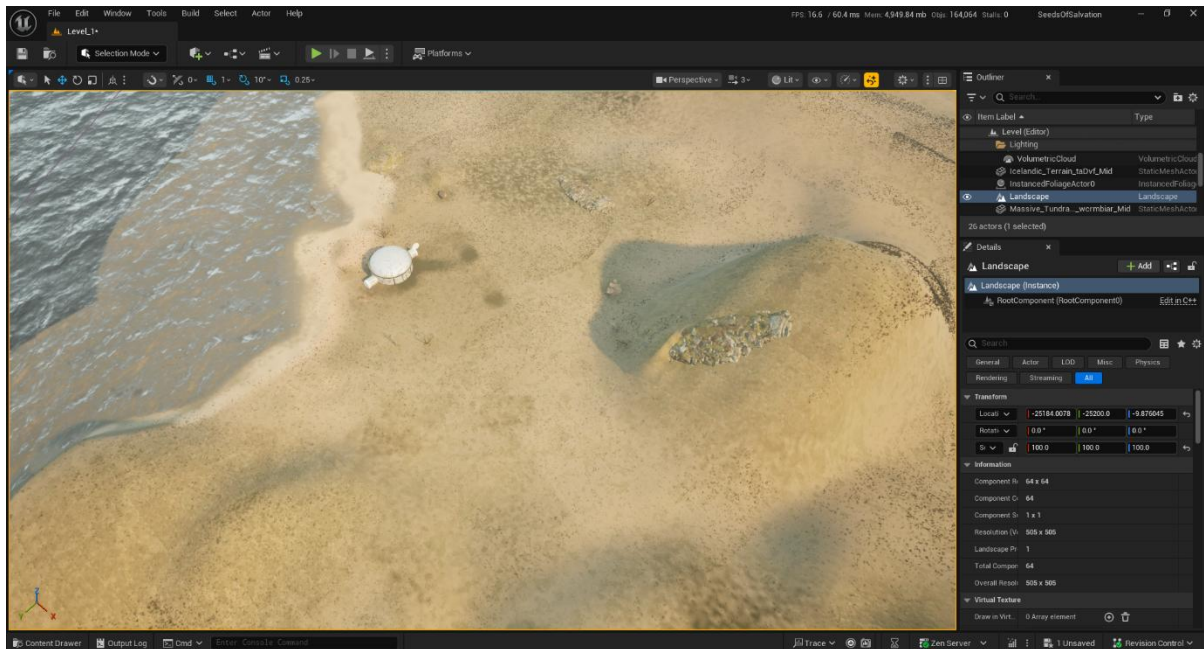


Figure 4.9: Map 2 (Top View)

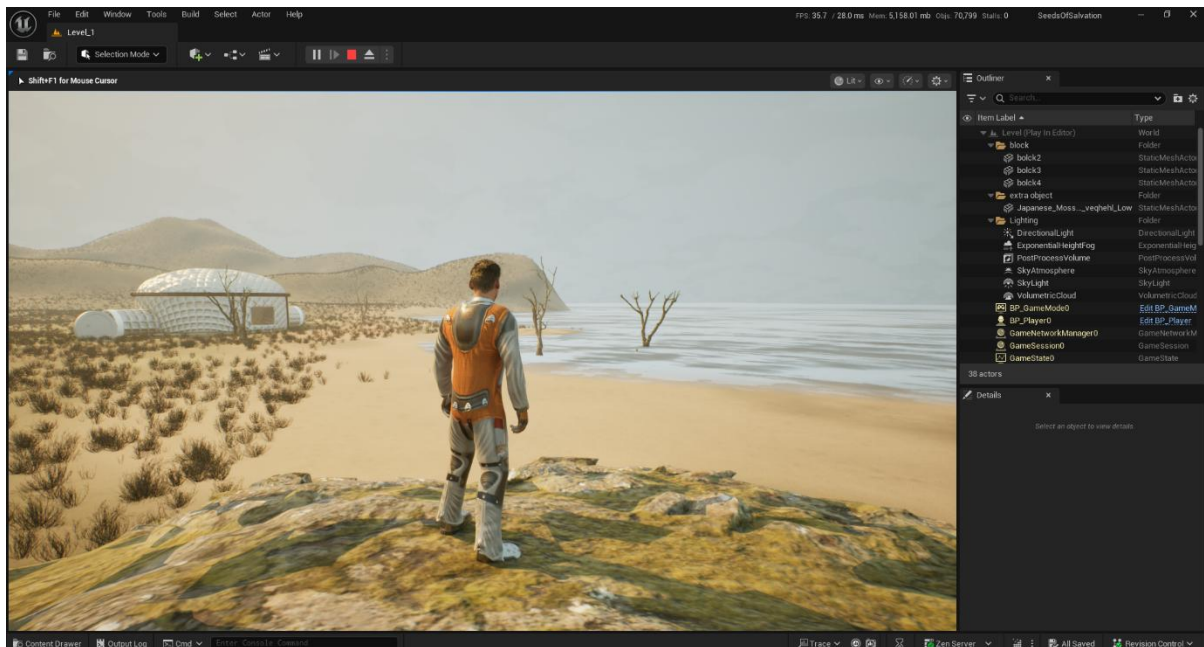


Figure 4.10: Map 2 (Environment)

- **Testing & Launch**

- xiii. Beta Release**

- In this stage, a beta version of the game was released to test.

- xiv. Feedback and Rebuild**

- After a beta released feedback was collected. After fixing cross device bug based on the feedback of the players, a new rebuild was released.

- xv. Deploy and Maintenance**

- This is a future stage. The new release will be deployed on the marketplace, and maintenance will start.

5.1 Limitations

Despite the potential impact of the project, two major category were identified during the development and evaluation phases. Those are:

- **Technical Limitation**

- i. Due to limited time, the implementation of advanced gameplay mechanic and realistic simulation were restricted.
- ii. Low-end devices affected the performance optimization.
- iii. For the limitation of enough human resource and skill like Sound Engineer, Graphics Designer, Motion Designer, VFX Artist, etc. resulted limitation in the simplification of graphics, animations, and AI systems.

- **Behavioral and Impact Limitation**

- i. Real-world behavioral change is not guaranteed despite having in-game awareness.
- ii. Individual interest and motivation have the most impacts in player engagement.
- iii. It cannot be controlled how seriously players take educational content.
- iv. Reach to age groups other than teenagers and young adult are limited due to the game focus.
- v. Non-gamers or individuals with limited access to digital devices are excluded.

5.2 Future Scope

The future scope of this project has significant opportunities for technical enhancement and educational expansion.

- **Technical Scope**

- i. Integration of AR(Augmented Reality) and VR (Virtual Reality) to provide ultra-realistic environmental simulations which will allow players to visualize the environmental effects in a more realistic manner.
- ii. Multiplayer and Community feature can be added to share eco-friendly ideas and strategies, development of community hub and cooperative modes.
- iii. Artificial Intelligence Integration to drive non-player (NPCs) and to personalize gameplay based on player decisions.

- **Educational Expansion Scope**

- i. Addition of city-specific scenarios with real-world environmental problems like air pollution, waste management, and water contamination.
- ii. Integration of real-life environmental data such as air quality indexes, water pollutions level, and climate statistics.
- iii. Collaboration with environmental organizations to use the game as an awareness, and further reach tool.
- iv. Use the game for campaigns, events and environmental challenges.

5.3 Conclusion

This project shows how video games can be used as an interactive medium to raise environmental awareness and inspire players' insights and attitudes toward real-world environmental problems. By combining storytelling, realistic challenges, and gameplay mechanics with scientifically justified environmental concepts, the project discloses both the issue of excessive digital engagement among teenagers and the lack of environmental awareness in society. The game offers a semi-real but hands-on experience that encourages learning through action and decision-making, unlike usual awareness campaigns. The project has certain limitations, but successfully shows that it is possible to use serious games as tools for education, and social change. Overall, this project focuses on the potential of game-based learning to inspire young minds to become responsible about the environment and motivate to contribute positively toward the sustainability of the environment around them.

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